



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Understanding TCP/IP & OSI Model

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Table of Contents

1. Questions and Answers.....	3
2. Practical Work	5

1. Questions and Answers

1. What is a network protocol?

- A network protocol is a set of rules that allows devices to communicate with each other over a network.

2. What is the OSI Model?

- The OSI Model is a seven-layer model that explains how data travels through a network.

3. Name all seven OSI layers in correct order.

- All seven OSI layers are:

- 7. Application Layer
- 6. Presentation Layer
- 5. Session Layer
- 4. Transport Layer
- 3. Network Layer
- 2. Data Link Layer
- 1. Physical Layer

4. What is the purpose of the Application Layer?

- It provides network services to applications such as web browser and email.

5. What is the purpose of the Transport Layer?

- It manages communication between devices and controls how data is delivered.

6. What is the purpose of the Network Layer?

- It handles IP addresses and decides the path for data to travel.

7. What is the purpose of the Data Link Layer?

- It handles communication within a local network using MAC addresses.

8. What is the purpose of the Physical Layer?

- It sends data as a signal through cables, Wi-Fi, or Fiber.

9. What is the TCP/IP Model?

- The TCP/IP Model is a networking model used for communication over the internet.

10. Explain the difference between OSI and TCP/IP models.

- OSI has 7 layers, while TCP/IP has 4 layers. OSI is mainly a reference model, while TCP/IP is widely used in real networks.

11. What is encapsulation?

- Encapsulation is a process of adding its header at each layer as data moves DOWN the stack.

12. What is decapsulation?

- Decapsulation is the process of removing the added data or header when the data moves UP the stack.

13. Which layer uses IP addresses?

- The Network Layer (Layer 3) uses IP addresses.

14. Which layer uses MAC addresses?

- The Data Link Layer (Layer 2) uses MAC addresses.

15. Which Layer uses TCP and UDP?

- The Transport Layer (Layer 4) uses TCP and UDP.

16. Which layer includes HTTP and HTTPS?

- The Application Layer (Layer 7) uses HTTP and HTTPS.

17. Why should cybersecurity professionals understand networking layers?

- It helps them understand network traffic, identify attacks, troubleshoot problems, and protect networks

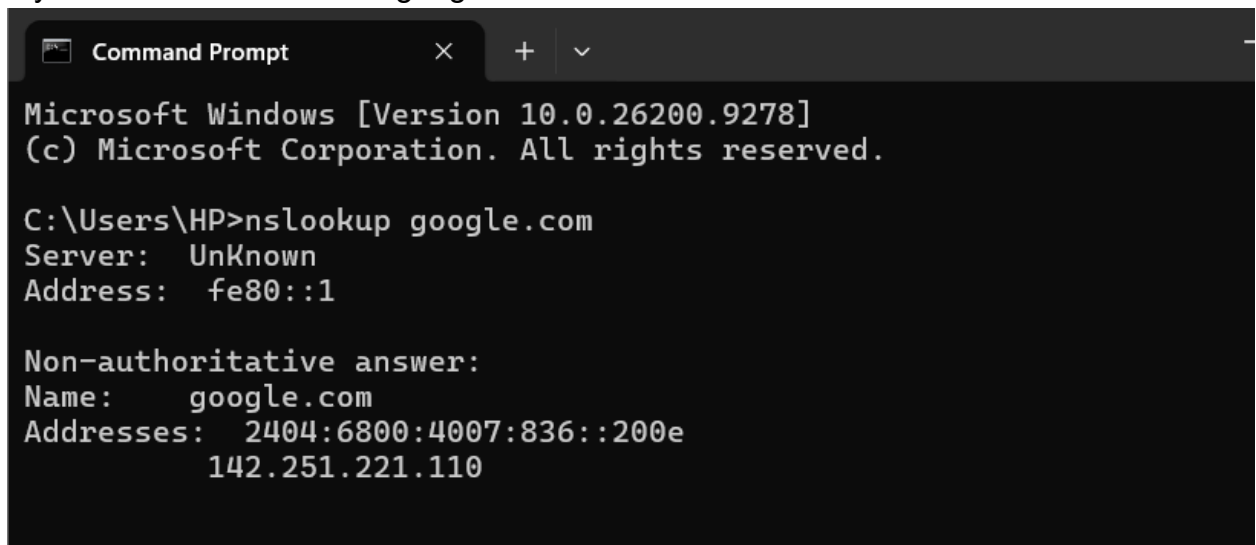
18. Explain how network data travels from a browser to server.

- The browser requests the website's IP address using DNS.
- The application creates the request, such as an HTTP/HTTPS request.
- The Transport Layer uses TCP (for typical HTTPS) to establish communication.
- The Network Layer adds the source and destination IP addresses.
- The Data Link Layer adds MAC address information and created frames.
- The Physical Layer sends the data as signals through the network.
- Router forward the packets to the server.
- The server receives the data, processes it and response back through the network using similar process.

2. Practical Work

19. Your resolved IP address for google.com (nslookup).

- My resolved IP address for google.com is 142.251.221.110



```
Command Prompt
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>nslookup google.com
Server:      UnKnown
Address:     fe80::1

Non-authoritative answer:
Name:       google.com
Addresses:  2404:6800:4007:836::200e
            142.251.221.110
```

20. Number of traceroute hops to google.com.

- Number of traceroute hops to google.com: 15 hops.

```
C:\Users\HP>tracert google.com

Tracing route to google.com [2404:6800:4007:836::200e]
over a maximum of 30 hops:

  1  141 ms    3 ms    1 ms    2404:7c00:48:a7ee:72a5:6aff:feb3:259a
  2   6 ms    4 ms    3 ms    2404:7c00::12
  3   9 ms    3 ms    3 ms    2404:7c00:0:d::1
  4  10 ms    3 ms    3 ms    2404:7c00:0:4007::1
  5  17 ms    3 ms    4 ms    2404:7c00:0:4002::1
  6 128 ms    7 ms    7 ms    2404:7c00:0:3::b
  7  27 ms    8 ms    7 ms    2404:a800:1a00:807::89
  8  77 ms   101 ms  101 ms  2404:a800::92
  9  69 ms   115 ms  112 ms  2404:6800:8202:200::1
 10 132 ms   101 ms  101 ms  2404:6800:8202:200::1
 11 115 ms   101 ms  101 ms  2001:4860:0:1::13dc
 12 112 ms   100 ms   57 ms  2001:4860:0:1::40e8
 13  *        151 ms  100 ms  2001:4860:0:1::8809
 14  62 ms   154 ms  101 ms  2001:4860:0:1::55d1
 15 109 ms   101 ms  100 ms  lcmaaa-az-in-x0e.1e100.net [2404:6800:4007:8
36::200e]

Trace complete.
```

21. Screenshot of Ping google.com



```
C:\Users\HP>ping google.com

Pinging google.com [2404:6800:4007:83d::200e] with 32 bytes of data:
Reply from 2404:6800:4007:83d::200e: time=144ms
Reply from 2404:6800:4007:83d::200e: time=61ms
Reply from 2404:6800:4007:83d::200e: time=76ms
Reply from 2404:6800:4007:83d::200e: time=95ms

Ping statistics for 2404:6800:4007:83d::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 61ms, Maximum = 144ms, Average = 94ms

C:\Users\HP>
```